

DLD HOMEWORK 4

10303

Q1:-

a)

X	Y	P(t)	P(t+1)
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

b)

XY \ P(t)	0	1
00	0	1
01	1	1
11	0	0
10	1	0

$$P(t+1) = X \bar{Y} \bar{P}(t) + \bar{X} Y + \bar{X} P(t)$$

c)

X	Y	P(t)	P(t+1)
0	0	0	0
1	1	0	0
0	1	0	1
1	0	0	1
1	X	1	0
0	X	1 0	1

c)

X	Y	Output. (P_{t+1})
0	0	$P(t)$
0	1	0 1
1	0	$P(t)'$
1	1	1 0

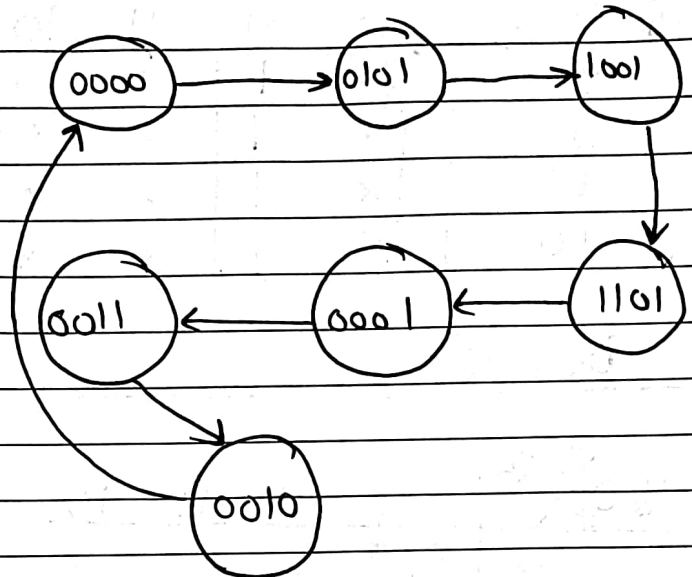
D)

A	B	P(t)	A+P(t)		B+P(t)	
			P(t) X	Y	P(t+1)	
0	0	0	0	0	0	
0	0	1	1	1	0	
0	1	0	0	1	1	
0	1	1	1	1	0	
1	0	0	1	0	1	
1	0	1	1	1	0	
1	1	0	1	1	0	
1	1	1	1	1	0	

Q2:-

a)

$0 = 0000$
 $5 = 0101$
 $9 = 1001$
 $13 = 1101$
 $1 = 0001$
 $3 = 0011$
 $2 = 0010$



B) Transition table:-

Previous state				Next state			
Q_3	Q_2	Q_1	Q_0	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0	1	0	1
0	1	0	1	1	0	0	1
1	0	0	1	1	1	0	1
1	1	0	1	0	0	0	1
0	0	0	1	0	0	1	1
0	0	1	1	0	0	1	0
0	0	1	0	⊗	⊗	⊗	⊗

c)

$D_0:-$ $Q_2 Q_1$	$Q_1 Q_0$	00	01	11	10
00		1	1	0	0
01		x	1	x	x
11		x	1	x	x
10		x	1	x	x

$$D_0 = \overline{Q_1}$$

$D_1:-$ $Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00		0	1	1	0
01		x	0	x	x
11		x	0	x	x
10		x	0	x	x

$$D_1 = \overline{Q_3} \overline{Q_2} Q_0$$

 $D_2:-$

$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00		1	0	0	0
01		x	0	x	x
11		x	0	x	x
10		x	1	x	x

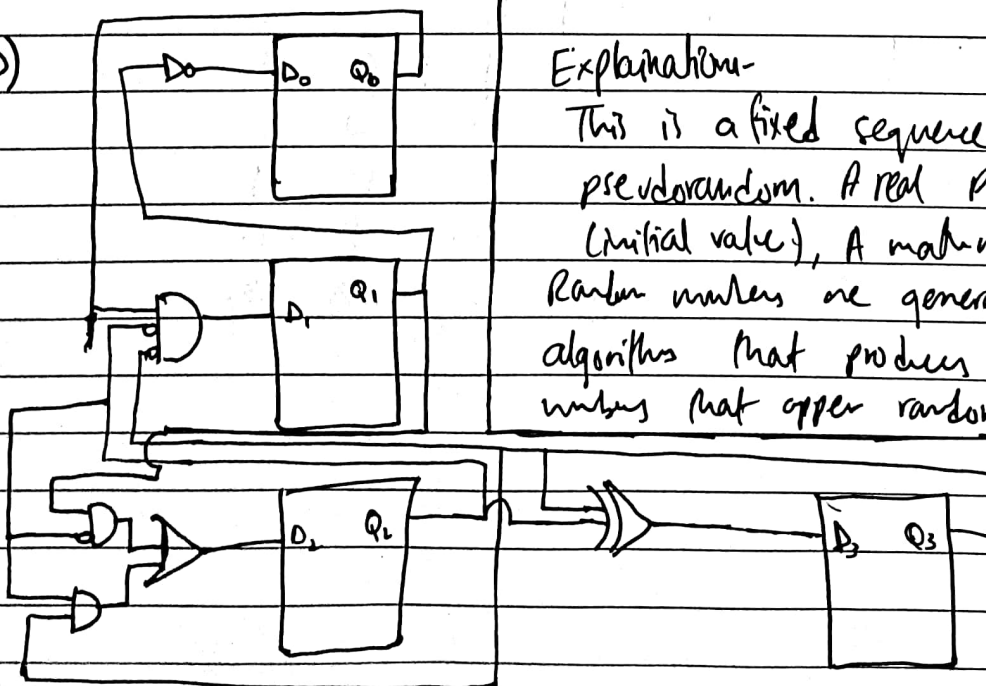
$$D_2 = \overline{Q_1} \overline{Q_2} + Q_3 \overline{Q_2}$$

 $D_3:-$

$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00		0	0	0	0
01		x	1	x	x
11		x	0	x	x
10		x	1	x	x

$$D_3 = \overline{Q_3} Q_2 + Q_3 \overline{Q_2}$$

D)



Explanation-

This is a fixed sequence generator not a pseudorandom. A real PRNG uses A seed (initial value), A mathematical algorithm. Random numbers are generated using deterministic algorithms that produce sequences of that values that appear random.

Q3:-

$$a) \quad Q_1^{(t+1)} = T_1 \oplus Q_1(t) \quad \text{--- eq(i)}$$

$$Q_2^{(t+1)} = T_2 \oplus Q_2(t) \quad \text{--- eq(ii)}$$

from the circuit:-

$$T_1 = X + \overline{Q_2}$$

$$T_2 = \overline{Q_1} + \overline{Q_2}$$

Now, put the

equation in (i) & (ii)

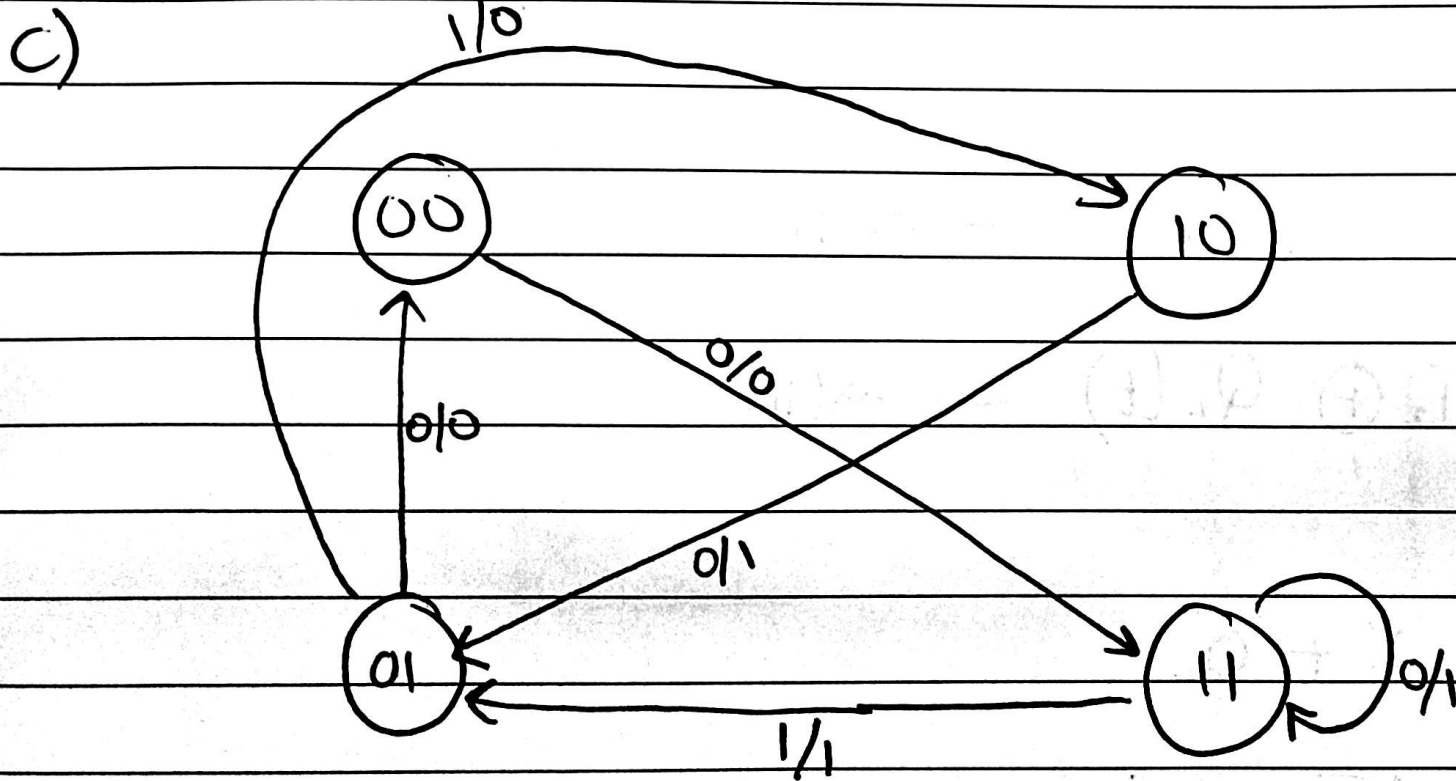
$$Q_1(t+1) = (X + \overline{Q_2}) \oplus Q_1(t)$$

$$Q_2(t+1) = (\overline{Q_1} + \overline{Q_2}) \oplus Q_2(t)$$

$$Y(\text{output}) = Q_1$$

b)	X	current states		Next states		T_1	T_2	Y
		$Q_1(t)$	$Q_2(t)$	$Q_1(t+1)$	$Q_2(t+1)$			
	0	0	0	1	1	1	1	0
	0	0	1	0	0	0	1	0
	0	1	0	0	1	1	1	1
	0	1	1	1	1	0	0	1
	1	0	0	1	1	1	1	0
	1	0	1	1	0	1	1	0
	1	1	0	0	1	1	1	1
	1	1	1	0	1	1	0	1

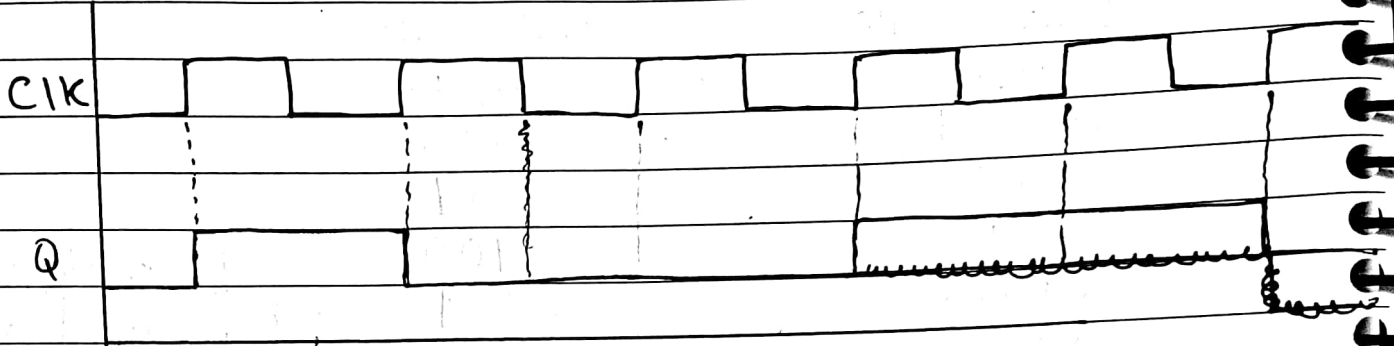
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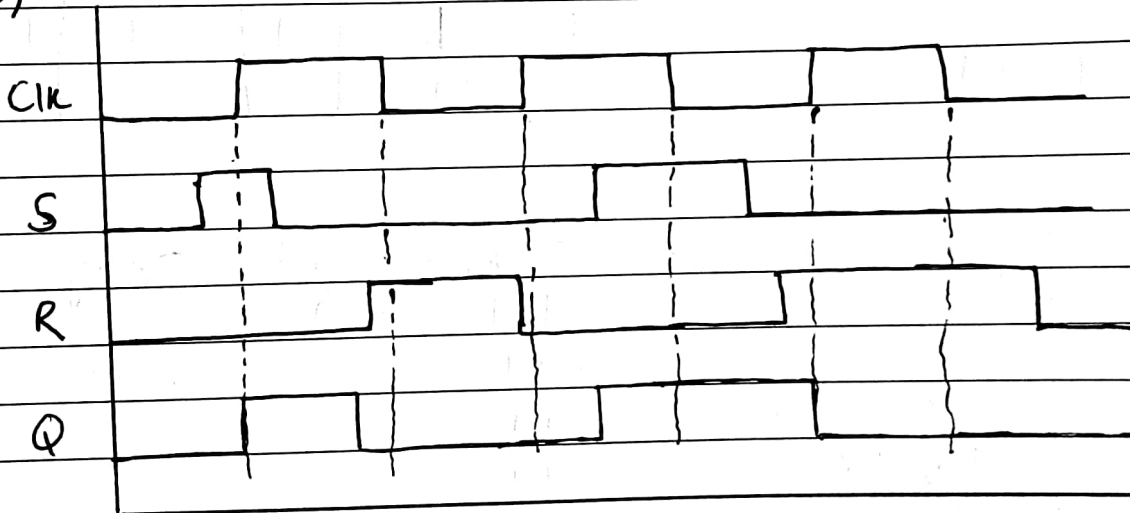
• As the output depends upon $Q_i(t)$ $\rightarrow$ It's a Moore's machine.

DLD HOMEWORK #4

Sualeh 10303

Q4
a)

b)

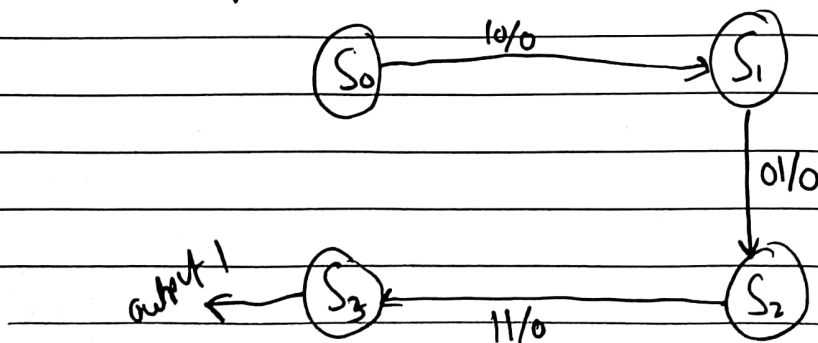


State table:-

- c) $S_0 = 00$ — idle
 $S_1 = 01$ — 'A' detected
 $S_2 = 10$ — 'B' detected
 $S_3 = 11$ — 'C' detected

Present state	inputs	Next state	Y (unlock)
S_0	00	S_0	0
S_0	10	S_1	0
S_0	01	S_0	0
S_0	11	S_2	0
S_1	00	S_1	0
S_1	10	S_1	0
S_1	01	S_2	0
S_1	11	S_2	0
S_2	00	S_2	0
S_2	10	S_2	0
S_2	01	S_2	0
S_2	11	S_3	0
S_3	00	S_3	1
S_3	10	S_0	0
S_3	01	S_0	0
S_3	11	S_0	0

State diagram:-



k-map for J_1

$Q_1 \backslash Q_0$		AB				Q
		00	01	11	10	
00		0	0	1	0	
01		0	0	1	0	
11		0	0	1	X	
10		0	0	1	X	

$$J_1 = \overline{Q_1} \overline{Q_0} AB$$

K map for k_{11} -

$Q_1 Q_0 \backslash AB$	00	01	11	10
00	X	X	X	X
01	X	X	X	X
11	0	0	0	0
10	0	1	1	1

$$k_1 = (\overline{Q_0} A) + (\overline{Q_0} B)$$

K map for J_{01} -

$Q_1 Q_0 \backslash AB$	00	01	11	10
00	0	0	1	1
01	X	X	X	X
11	X	X	0	X
10	0	0	0	0

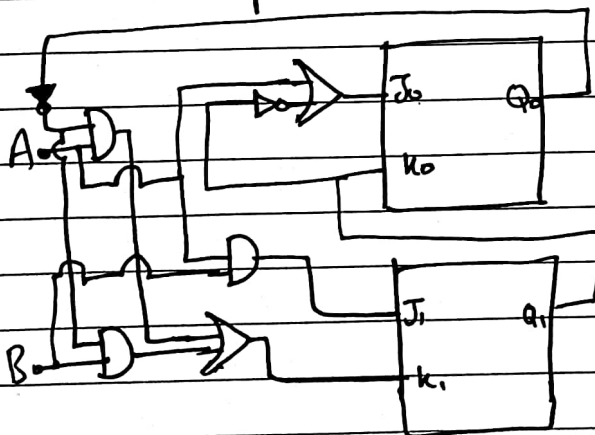
$$J_0 = \overline{Q_1} + A$$

K map for k_{01} -

$Q_1 Q_0 \backslash AB$	00	01	11	10
00	X	X	X	X
01	0	0	0	0
11	X	X	1	X
10	X	X	X	X

$$k_0 = Q_1$$

Circuit diagram



b)

- As only we can see that $b \neq e$ have the same output and next states hence we can say that $b=e$.
- As $b=e$ so a also equals to c because their output and next state will also be same.
- $d=h$, because output and next states are same.

d)

Reduced table:-

Present state	$X=0$		$X=1$	
	Next state	Output	Next state	Output
a	f	0	b	0
b	d	0	a	0
d	g	1	a	0
f	f	1	b	1
g	g	0	d	1

Reduced state diagram:-

